H524 Week 4 Assignment: Confidence Intervals

Questions with Answer Key

Introduction to Biostatistics

Fall 2025

Assignment Information:

- **Total Points:** 110 points
- **Time Limit:** 180 minutes (3 hours)
- **Attempts:** 1
- **Question Types:** 16 Multiple Choice (80 pts), 2 Essay/Fill-in (30 pts)

Part 1: CI Fundamentals

Question 1: CI Interpretation (5 points)

A study reports a 95% confidence interval for mean blood pressure reduction as (8.2, 14.6) mmHg. What is the CORRECT interpretation?

a) We are 95% confident that the true population mean blood pressure reduction is between 8.2 and 14.6 mmHg

b) There is a 95% probability that the true mean is between 8.2 and 14.6 mmHg

c) 95% of patients will have blood pressure reductions between 8.2 and 14.6 mmHg

d) The sample mean has a 95% chance of being between 8.2 and 14.6 mmHg

ANSWER: a) We are 95% confident that the true population mean blood pressure reduction is between 8.2 and 14.6 mmHg.

Explanation: This is the correct frequentist interpretation. A confidence interval describes our confidence about the location of the population parameter (the true mean), not the probability that the parameter falls in the interval (it either does or doesn't), and not the distribution of individual values.

Question 2: Critical Value Selection (5 points)

You want to construct a 95% confidence interval for a population mean. The sample size is $n = 15$, and the population standard deviation is unknown. Which critical value should you use?

a) $z = 1.96$

b) $t(14) = 2.145$

c) $t(15) = 2.131$

d) $z = 2.576$

ANSWER: b) $t(14) = 2.145$

Explanation: Since the population standard deviation (σ) is unknown and we're using the sample standard deviation (s), we must use the t-distribution. The degrees of freedom are $df = n - 1 = 15 - 1 = 14$. For a 95% CI, we use $t_{0.025,14} = 2.145$.

Question 3: Factors Affecting CI Width (5 points)

If you want a NARROWER confidence interval for a population mean, which of the following actions would achieve this?

a) Increase the confidence level from 95% to 99%

b) Decrease the sample size

c) Increase the sample size

d) Use a larger standard deviation

ANSWER: c) Increase the sample size

Explanation: The margin of error is $ME = t_{\alpha/2} \times \frac{s}{\sqrt{n}}$. Increasing n decreases the standard error ($s/\sqrt{n}$), which narrows the CI. Options a, b, and d would all make the CI wider.

Question 4: Standard Error (5 points)

A study has a sample standard deviation $s = 12$ and sample size $n = 36$. What is the standard error of the mean?

a) $SE = 2$

b) $SE = 3$

c) $SE = 6$

d) $SE = 12$

ANSWER: a) $SE = 2$

Calculation: $SE = \frac{s}{\sqrt{n}} = \frac{12}{\sqrt{36}} = \frac{12}{6} = 2$

Part 2: CI for Means

Question 5: Manual CI Calculation (15 points)

A clinical trial measures pain reduction scores for 25 patients receiving a new treatment. The results are:

- Sample mean: $\bar{x} = 7.8$ points
- Sample standard deviation: $s = 3.2$ points
- Sample size: $n = 25$

Calculate a 95% confidence interval for the true mean pain reduction. Show all work including:

a) Which distribution to use (z or t) and why (3 points)

b) The appropriate critical value (3 points)

c) The standard error calculation (3 points)

d) The margin of error calculation (3 points)

e) The final confidence interval with interpretation (3 points)

Hint: For a 95% CI with df=24, the critical t-value is approximately 2.064

ANSWER: Complete Solution:

a) Distribution choice (3 points): Use the t-distribution because:

- The population standard deviation (σ) is unknown
- We're using the sample standard deviation ($s = 3.2$)
- Even though $n = 25$ is relatively small, the t-distribution is the appropriate choice

b) Critical value (3 points):

- Degrees of freedom: $df = n - 1 = 25 - 1 = 24$
- For 95% CI: $t_{0.025,24} = 2.064$

c) Standard error (3 points):

$$SE = \frac{s}{\sqrt{n}} = \frac{3.2}{\sqrt{25}} = \frac{3.2}{5} = 0.64$$

d) Margin of error (3 points):

$$ME = t_{\alpha/2} \times SE = 2.064 \times 0.64 = 1.32$$

e) Final CI and interpretation (3 points):

$$CI = \bar{x} \pm ME = 7.8 \pm 1.32 = (6.48, 9.12)$$

Interpretation: We are 95% confident that the true mean pain reduction for all patients receiving this treatment is between 6.48 and 9.12 points.

Question 6: Confidence Level Comparison (5 points)

Using the same data from the Manual CI question above, if we constructed a 99% confidence interval instead of 95%, what would happen to the interval width?

a) The interval would be narrower because we are more confident

b) The interval would be wider because higher confidence requires a larger margin of error

c) The interval width would stay exactly the same

d) The interval would be narrower because the critical value decreases

ANSWER: b) The interval would be wider because higher confidence requires a larger margin of error

Explanation: To be more confident (99% vs. 95%), we need to cast a wider net. The critical value increases from $t_{0.025,24} = 2.064$ to $t_{0.005,24} \approx 2.797$, which increases the margin of error and widens the interval.

Part 3: CI for Proportions

Question 7: Proportion CI Conditions (5 points)

When constructing a confidence interval for a population proportion using the normal approximation (Wald method), which condition must be satisfied?

a) $n \geq 30$

b) Both $np \geq 5$ and $n(1-p) \geq 5$

c) The population must be normally distributed

d) The sample proportion must be greater than 0.5

ANSWER: b) Both $np \geq 5$ and $n(1-p) \geq 5$

Explanation: The normal approximation to the binomial distribution requires both the expected number of successes (np) and failures ($n(1-p)$) to be at least 5. This ensures the sampling distribution of $\hat{p}$ is approximately normal.

Question 8: Proportion CI Calculation (5 points)

In a survey of 200 patients, 150 reported satisfaction with their treatment. What is the sample proportion ($\hat{p}$)?

a) $\hat{p} = 0.50$

b) $\hat{p} = 0.67$

c) $\hat{p} = 0.75$

d) $\hat{p} = 0.80$

ANSWER: c) $\hat{p} = 0.75$

Calculation: $\hat{p} = \frac{x}{n} = \frac{150}{200} = 0.75$

Question 9: Standard Error for Proportion (5 points)

Using the data from the proportion calculation question ($\hat{p} = 0.75$, $n = 200$), what is the standard error for the proportion? Use $SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

a) $SE \approx 0.031$

b) $SE \approx 0.043$

c) $SE \approx 0.075$

d) $SE \approx 0.125$

ANSWER: a) $SE \approx 0.031$

Calculation:

$$SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.75 \times 0.25}{200}} = \sqrt{\frac{0.1875}{200}} = \sqrt{0.0009375} \approx 0.0306 \approx 0.031$$

Question 10: Vaccine Efficacy CI (5 points)

A vaccine trial reports a 95% CI for vaccine efficacy as (72%, 89%). What can we conclude?

a) We are 95% confident the true efficacy is between 72% and 89%, suggesting the vaccine is effective since the entire interval is above 0%

b) 72% of people are protected at minimum, 89% at maximum

c) The vaccine has a 95% chance of working

d) We cannot determine effectiveness because the interval is too wide

ANSWER: a) We are 95% confident the true efficacy is between 72% and 89%, suggesting the vaccine is effective since the entire interval is above 0%

Explanation: This is the correct interpretation. Since the entire interval is above 0%, we have evidence that the vaccine is effective. If the interval included 0%, we couldn't conclude effectiveness.

Part 4: Sample Size Determination

Question 11: Sample Size Concepts (5 points)

You want to estimate the mean cholesterol level in a population with a margin of error of 5 mg/dL at 95% confidence. If you quadruple the sample size, what happens to the margin of error?

a) The margin of error is cut in half

b) The margin of error is cut to one-quarter

c) The margin of error doubles

d) The margin of error stays the same

ANSWER: a) The margin of error is cut in half

Explanation: Since $ME \propto \frac{1}{\sqrt{n}}$, quadrupling the sample size ($n \to 4n$) means:

$$ME_{new} = \frac{ME_{old}}{\sqrt{4}} = \frac{ME_{old}}{2}$$

The margin of error decreases by a factor of $\sqrt{4} = 2$.

Question 12: Sample Size Calculation (5 points)

A researcher wants to estimate a population proportion with a 95% CI and margin of error of 0.05. Using the conservative estimate ($\hat{p} = 0.5$), what sample size is needed?

Formula: $n = \frac{z^2 \cdot \hat{p}(1-\hat{p})}{E^2}$, where $z = 1.96$ for 95% CI

a) $n = 196$

b) $n = 271$

c) $n = 385$

d) $n = 577$

ANSWER: c) $n = 385$

Calculation:

$$n = \frac{z^2 \cdot \hat{p}(1-\hat{p})}{E^2} = \frac{(1.96)^2 \cdot 0.5 \cdot 0.5}{(0.05)^2} = \frac{3.8416 \cdot 0.25}{0.0025} = \frac{0.9604}{0.0025} = 384.16 \approx 385$$

Always round up for sample size calculations.

Part 5: Assumptions and Checking

Question 13: Normality Assessment (5 points)

You have a sample of $n = 12$ observations and want to construct a t-based confidence interval for the mean. Which of the following is the BEST way to assess whether the normality assumption is reasonable?

a) Check that $n \geq 30$, which automatically ensures normality

b) Create a histogram and Q-Q plot to visually assess normality, and consider the Shapiro-Wilk test

c) Calculate the mean and median; if they're close, the data is normal

d) Normality is not required for t-based intervals

ANSWER: b) Create a histogram and Q-Q plot to visually assess normality, and consider the Shapiro-Wilk test

Explanation: With small samples ($n < 30$), visual inspection (histogram, Q-Q plot) combined with formal tests (Shapiro-Wilk) is the best approach. Option a is wrong because $n \geq 30$ doesn't guarantee normality (it just helps by CLT). Option c is insufficient. Option d is wrong because t-intervals assume either normality or large n.

Question 14: Large vs Small Sample (5 points)

When can you safely use a z-based confidence interval instead of a t-based confidence interval for a population mean?

a) When the population standard deviation (σ) is known, regardless of sample size

b) Only when $n \geq 30$

c) When the sample size is less than 30

d) Never; always use t-distribution for real-world data

ANSWER: a) When the population standard deviation (σ) is known, regardless of sample size

Explanation: If σ is truly known (rare in practice), you can use the z-distribution regardless of sample size. In practice, since σ is almost never known, we use t. The "$n \geq 30$" rule is a rough guideline for when t and z are similar, but t is still more appropriate when using s.

Question 15: CI for Hypothesis Testing (5 points)

A 95% confidence interval for the difference in mean blood pressure between two treatments is $(-2.3, 8.7)$ mmHg. What can you conclude at the $\alpha = 0.05$ significance level?

a) There is a significant difference between treatments because the interval is positive

b) There is NOT a significant difference between treatments because the interval contains 0

c) Treatment 1 is better than Treatment 2

d) We need more data to make any conclusion

ANSWER: b) There is NOT a significant difference between treatments because the interval contains 0

Explanation: A 95% CI corresponds to a two-sided test at $\alpha = 0.05$. Since the interval contains 0 (the null hypothesis value for "no difference"), we fail to reject the null hypothesis. There is insufficient evidence of a significant difference between treatments.

Part 6: R Implementation

Question 16: CI with R (15 points)

A clinical trial measures vitamin D levels (ng/mL) for 30 patients after supplementation:

```
vitamin_d <- c(32, 38, 29, 35, 31, 40, 27, 34, 36, 30,
               28, 37, 33, 35, 31, 34, 36, 29, 33, 37,
               30, 35, 34, 31, 39, 33, 36, 29, 34, 35)
```

Using R, complete the following tasks and paste your code and results:

a) Calculate the sample mean, standard deviation, and sample size (3 points)

b) Create a histogram to assess normality (2 points)

c) Use `t.test()` to calculate a 95% confidence interval for the mean vitamin D level (5 points)

d) Interpret the confidence interval in context (3 points)

e) What would you conclude about whether the mean vitamin D level is above 30 ng/mL? (2 points)

ANSWER: Complete R Solution:

a) Descriptive statistics (3 points):

```
vitamin_d <- c(32, 38, 29, 35, 31, 40, 27, 34, 36, 30,
               28, 37, 33, 35, 31, 34, 36, 29, 33, 37,
               30, 35, 34, 31, 39, 33, 36, 29, 34, 35)

# Sample statistics
mean(vitamin_d)
# [1] 33.37

sd(vitamin_d)
# [1] 3.35

length(vitamin_d)
# [1] 30
```

b) Histogram for normality (2 points):

```
hist(vitamin_d,
     main = "Histogram␣of␣Vitamin␣D␣Levels",
     xlab = "Vitamin␣D␣(ng/mL)",
     col = "lightblue",
     breaks = 10)
```

The histogram should show approximately normal distribution.

c) 95% CI using t.test() (5 points):

```
t.test(vitamin_d, conf.level = 0.95)

# Output:
# One Sample t-test
#
# data:  vitamin_d
# t = 54.595, df = 29, p-value < 2.2e-16
# alternative hypothesis: true mean is not equal to 0
# 95 percent confidence interval:
#  32.11669 34.61664
# sample estimates:
# mean of x
#    33.36667
```

d) Interpretation (3 points):

We are 95% confident that the true mean vitamin D level for all patients receiving this supplementation is between 32.12 and 34.62 ng/mL.

e) Conclusion about 30 ng/mL (2 points):

Since the entire 95% confidence interval (32.12, 34.62) is above 30 ng/mL, we can conclude with 95% confidence that the true mean vitamin D level is greater than 30 ng/mL. This suggests the supplementation successfully raises vitamin D levels above the 30 ng/mL threshold.

Question 17: Proportion CI in R (5 points)

Which R function would you use to calculate a confidence interval for a population proportion?

a) `prop.test(x, n)`

b) `t.test(x)`

c) `binom.test(x, n)`

d) Both a and c are appropriate

ANSWER: d) Both a and c are appropriate

Explanation:

- `prop.test(x, n)` uses the normal approximation (Wald method) and is appropriate for large samples where $np \geq 5$ and $n(1-p) \geq 5$
- `binom.test(x, n)` uses the exact binomial method and is appropriate for all sample sizes, especially small samples
- Both provide confidence intervals for proportions, with different methods

Summary of Answers

Q	**Answer**	**Q**	**Answer**	**Q**	**Answer**
1	a	7	b	13	b
2	b	8	c	14	a
3	c	9	a	15	b
4	a	10	a	17	d
6	b	11	a		
		12	c		

Note: Questions 5 and 16 are essay/fill-in questions requiring detailed written responses (see full answers above).